

Guarded Current-Evidence Fusion for Cooperative Multitarget Tracking

Abstract—Cooperative trackers exchange posteriors that mix inherited information with current observations. Weighting these posteriors by age cannot control the two contributions separately. We propose guarded current-evidence fusion (GCE), which augments a conservative posterior pool with bounded Bernoulli posterior-to-prediction ratios. Each admitted ratio jointly updates the spatial density and existence probability through a shared normalizer. A Gaussian curvature guard makes this correction tractable under approximate association, and an exact zero-vector codec reduces its communication cost. On 25 V2V4Real sequences with a common score-aware local tracker, GCE reduces OSPA by 6.7% and 5.7% relative to qualified geometric pooling under reliable and intermittent links. It also improves on recency weighting and scalar-only correction in both conditions. The gain combines lower missed-target and false-target costs with better common-target localization. Exact zero omission cuts fragmented communication by 11.3% and 10.8% relative to full Gaussian packets while preserving the tracking outputs. These results demonstrate the benefit of restoring current evidence jointly across the two parts of a Bernoulli belief.

I. INTRODUCTION

Cooperative robots observe different targets, occlusions, and missed detections. Their exchanged tracks also contain information inherited from earlier communication. A recently received posterior may therefore carry old evidence, while a low existence probability may convey a useful current miss. Effective fusion must distinguish inherited beliefs from new observations without transmitting the entire sensing history.

Labeled multi-Bernoulli (LMB) filtering represents each track by a label, existence probability, and conditional spatial density [1]. Geometric pooling, or Kullback–Leibler averaging (KLA), combines local posteriors when their cross-correlations are unavailable [2]. In the ideal shared-prior case, normalized geometric weights retain one copy of the prior but also temper each new likelihood. Changing posterior weights according to observation age does not separately control these contributions. A further coupling arises within each Bernoulli: the integral of its spatial product affects the probability that the target exists. An existence correction alone leaves this part of the current evidence unused.

We propose *guarded current-evidence fusion* (GCE). Each robot encodes the ratio between its local update and prediction. Fusion augments a conservative posterior pool with admitted fractions of these ratios, then jointly normalizes existence and spatial density. Current sensing opportunity, association-weighted score support, and Gaussian curvature determine which increments are admitted. Figure 1 shows how the same current evidence reaches both branches of the fused belief.

Prior–likelihood separation is established [3], [4]. GCE develops this principle into a bounded Bernoulli correction for approximate local tracking with unequal sensing histories. The paper contributes:

- A joint existence and spatial correction with closed-form Gaussian normalization and a source-level curvature guard.
- An admission rule that combines current association, detection-score, and missed-detection evidence, plus an exact codec for zero increments.
- A cooperative tracking evaluation with common local filtering, geometric and arithmetic fusion references, component ablations, and measured packet sizes.

Across both link conditions, GCE achieves the lowest main-comparison OSPA. Relative to no-age pooling and recency weighting, missed-target cost, false-target cost, and common-target localization improve together.

II. RELATED WORK

A. Cooperative robotic tracking

Distributed PHD filtering supports robot search and tracking [5], and multirobot reconfiguration adapts sensing resources [6]. LMB representations also support distributed landmark mapping [7]. For cooperative driving, V2V4Real provides real multi-vehicle observations [8], while DM-STrack learns uncertainty in a differentiable multisensor tracker [9]. We use its released per-vehicle detections for planar LMB tracking and focus on the fusion layer.

B. Finite-set fusion and sensing support

Different fields of view require explicit treatment of absent labels and exclusive information [10], [11]. Minimum-information-loss (MIL) fusion uses arithmetic pooling of labeled finite-set densities [12]; track consensus matches and combines short trajectories [13]. Our comparisons adapt the public MIL author manuscript [14] and the author’s track-consensus functions to the common detections.

Cardinality and localization also play distinct roles in finite-set fusion: pointwise density consistency alone does not ensure a useful cardinality distribution [15]. Heterogeneous fusion assigns confidence at the component level [16]. GCE instead couples the two components through the normalizer of an admitted Bernoulli update.

C. Prior and current likelihood information

Shared-prior Bayesian fusion corrects for common information in local posteriors [3]. Likelihood consensus aggregates exponential-family likelihood representations [4]. GCE

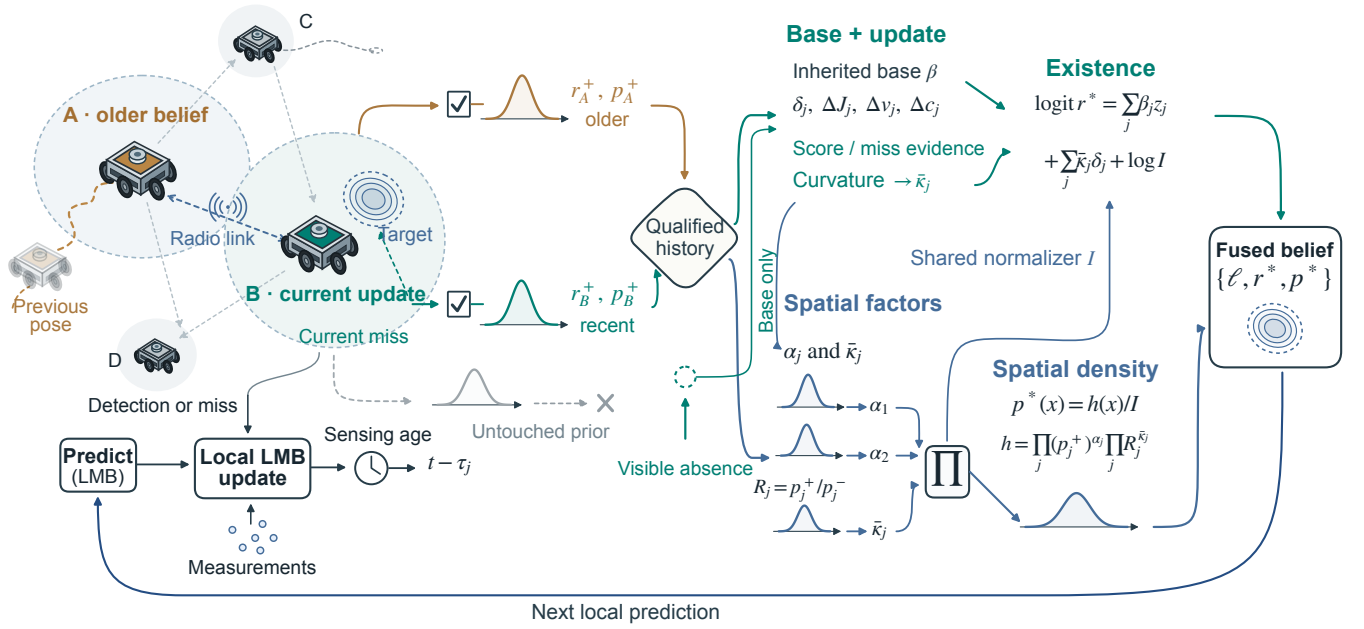


Fig. 1. Guarded current-evidence fusion. Robots exchange inherited beliefs and current Bernoulli ratios. Qualified support forms the base; sensing, score, and curvature determine the admitted correction. The same ratio updates spatial density and existence through the shared integral I . The fused belief becomes the next local prior. Robot poses and density shapes illustrate the mechanism.

transmits Gaussian natural-parameter increments and admits them fractionally. This accommodates local LMB association and moment projection, for which the posterior-to-prediction ratio can differ from an exact sensor likelihood.

III. GUARDED CURRENT-EVIDENCE FUSION

A. Local prediction, update, and information support

Each robot predicts its LMB, updates it with the current local measurement set, and exchanges one posterior packet with available neighbors. Fusion output becomes the next local prior. A label is aligned across sources before fusion; births remain local and measurement driven. For a represented label, write the local prediction and update as $f_j^- = (r_j^-, p_j^-)$ and $f_j^+ = (r_j^+, p_j^+)$. The Bernoulli density has empty-set mass $1-r$ and singleton density $rp(x)$. The implementation moment-projects p_j^- and p_j^+ to four-dimensional Gaussians over planar position and velocity.

A direct sensing opportunity occurs when the modeled detection probability at a predicted track is positive. Both a detection and a missed detection advance the local direct-opportunity timestamp τ_j ; receiving another robot's packet does not advance it. A separate history flag records whether a represented belief descends from an informative local update. When informed inputs exist, untouched priors are excluded. An absent label contributes an existence censor only if its source could observe the represented target. Otherwise that source abstains. Censors carry no spatial density. This support handling is shared with the qualified no-age and recency controls.

Let $\mathcal{P}$ be eligible represented sources and $\mathcal{E}$ eligible existence sources, including qualified censors. Normalize or-

dinary delivered weights w_j separately on the two supports:

$$\alpha_j = \frac{w_j \mathbf{1}[j \in \mathcal{P}]}{\sum_{k \in \mathcal{P}} w_k}, \quad b_j = \frac{w_j \mathbf{1}[j \in \mathcal{E}]}{\sum_{k \in \mathcal{E}} w_k}. \quad (1)$$

All sums below use the corresponding support. A label without spatial support is omitted. For a represented source, define

$$a_j = \rho + (1 - \rho)e^{-(t - \tau_j)\Delta t / \lambda}, \quad q_j = \frac{b_j a_j}{\sum_k b_k a_k}, \quad (2)$$

with $a_j = \rho$ before any local opportunity and $a_j = 1$ for a censor.

B. A conservative inherited base

Set $z_j = \text{logit}(r_j^+)$ and choose the existence weights

$$\beta = \begin{cases} q, & \sum_j (q_j - b_j) z_j < 0, \\ b, & \text{otherwise.} \end{cases} \quad (3)$$

Thus history can lower the base existence log-odds but cannot raise them. The spatial base retains the ordinary weights α . An unnormalized base has masses

$$A_0 = \prod_j (1 - r_j^+)^{\beta_j}, \quad (4)$$

$$A_1(x) = \prod_j (r_j^+)^{\beta_j} \prod_j p_j^+(x)^{\alpha_j}. \quad (5)$$

These are separate existence and spatial weights; when the weights and supports coincide, the base reduces to ordinary Bernoulli KLA. The no-history ablation sets $\beta = b$.

C. Admitting current positive and negative evidence

The local existence increment is

$$\delta_j = \text{logit}(r_j^+) - \text{logit}(r_j^-). \quad (6)$$

The local tracker incorporates a detection score s through a fitted positive-to-negative mark ratio $L(s) > 0$. Let W_{jm} be the normalized local association mass for assigning this Bernoulli to measurement m , and let $A_j = \sum_m W_{jm}$. For a current sensing opportunity, use

$$g_j^+ = \sum_m W_{jm} \max\left(0, \frac{L(s_m) - 1}{L(s_m) + 1}\right), \quad (7)$$

$$g_j^- = (1 - A_j) \frac{p_{D,j}}{2 - p_{D,j}}. \quad (8)$$

Both gates are zero without a current opportunity. The first admits positive increments in proportion to association-weighted mark discrimination; the second weights the missed-detection branch by its bounded likelihood contrast. They satisfy $g_j^+ \leq A_j$, $g_j^- \leq 1 - A_j$, and hence $g_j^+ + g_j^- \leq 1$.

If at least two eligible sources represent the label and none of the eligible existence inputs is a censor, define

$$\kappa_j = (1 - \beta_j) \begin{cases} g_j^+, & \delta_j \geq 0, \\ g_j^-, & \delta_j < 0. \end{cases} \quad (9)$$

Otherwise set every $\kappa_j = 0$. This restriction avoids constructing an unavailable current ratio for an absent label. The no-score ablation replaces the contrast in (7) by one, retaining the same score-aware local tracker and setting $g_j^+ = A_j$.

D. Joint existence and spatial normalization

For each admitted source, multiply the same fractional Bernoulli ratio $(f_j^+/f_j^-)^{\kappa_j}$ into both hypotheses in (5). The resulting spatial product is

$$h(x) = \prod_j p_j^+(x)^{\alpha_j} \prod_j \left[\frac{p_j^+(x)}{p_j^-(x)} \right]^{\kappa_j}, \quad I = \int h(x) dx. \quad (10)$$

Proposition 1 (normalized Bernoulli form). For $0 < r_j^-, r_j^+ < 1$ and $0 < I < \infty$, normalizing these empty and singleton masses yields

$$p^*(x) = h(x)/I, \quad (11)$$

$$\text{logit}(r^*) = \sum_j \beta_j z_j + \sum_j \kappa_j \delta_j + \log I. \quad (12)$$

Proof. The empty mass is $H_0 = A_0 \prod_j [(1 - r_j^+)/(1 - r_j^-)]^{\kappa_j}$. The integrated singleton mass is $H_1 = I \prod_j (r_j^+)^{\beta_j} \prod_j (r_j^+/r_j^-)^{\kappa_j}$. Taking $\log(H_1/H_0)$ gives (12); normalizing the singleton shape gives (11). $\square$

A scalar-only correction keeps the first two terms in (12) but uses the old spatial pool and its integral. GCE instead recomputes the spatial density and its normalizer from the admitted ratios, retaining their coupled effect on the Bernoulli belief.

In the special case of a common prior f^- , equal weights $\alpha = \beta$, unit gates, and independent exact local observations, $\kappa_j = 1 - \beta_j$. The full product then reduces to

$\prod_j f_j^+ / (f^-)^{M-1}$, the standard shared-prior Bayesian combination [3].

E. Gaussian correction and curvature guard

For $p = \mathcal{N}(m, P)$, write

$$\log p(x) = -\frac{1}{2} x^\top J x + v^\top x + c, \quad J = P^{-1}, \quad v = Jm, \quad (13)$$

where $c = -[d \log(2\pi) + \log |P| + m^\top Jm]/2$ and $d = 4$. A source transmits the differences $(\Delta J_j, \Delta v_j, \Delta c_j)$ between its update and prediction. The four-dimensional Gaussian ratio requires fifteen numbers: ten for the symmetric precision, four for the information vector, and one constant.

Moment projection can widen a covariance after ambiguous association, so this Gaussian ratio need not be log-concave. GCE admits a source correction only when its precision increment is positive semidefinite, up to numerical tolerance:

$$\bar{\kappa}_j = \kappa_j \mathbf{1}[\Delta J_j \succeq 0]. \quad (14)$$

The guard acts on the entire ratio, including the scalar increment. The aggregate natural parameters are

$$J = \sum_j \alpha_j J_j^+ + \sum_j \bar{\kappa}_j \Delta J_j, \quad (15)$$

$$v = \sum_j \alpha_j v_j^+ + \sum_j \bar{\kappa}_j \Delta v_j, \quad (16)$$

$$c = \sum_j \alpha_j c_j^+ + \sum_j \bar{\kappa}_j \Delta c_j. \quad (17)$$

For positive definite J , the normalized quantities are

$$P^* = J^{-1}, \quad m^* = J^{-1}v, \quad (18)$$

$$\log I = c + \frac{1}{2} [d \log(2\pi) - \log |J| + v^\top J^{-1}v]. \quad (19)$$

Use $\bar{\kappa}$ in (12) as well. A nonfinite or ill-conditioned aggregate, or a failed Cholesky factorization, restores the base. In exact arithmetic, positive semidefinite admitted precision gains and a positive definite base ensure a positive definite aggregate. The no-curvature ablation removes the source test while retaining the aggregate integrability check. Current ratios and gates are recomputed after each local update. For state dimension d and M eligible sources, the dense Gaussian arithmetic costs $O(Md^3)$ and adds $O(d^2)$ numbers per track.

F. Exact zero-vector transmission

The full Gaussian implementation extends a 232-byte scalar Bernoulli summary by fifteen 64-bit numbers, giving 352 bytes per track plus a 32-byte packet header. Many local updates leave all fifteen differences exactly zero. A two-byte tag distinguishes a full vector from an omitted zero vector and preserves the sign bit of each omitted zero. All nonzero vectors retain their original 120 bytes. For n tracks and k nonzero vectors, the actual packet length is

$$B = 32 + 234n + 120k \quad \text{bytes}. \quad (20)$$

This codec has no threshold or quantization parameter. Decoded numerical values, and hence the fusion rule, are unchanged.

IV. EXPERIMENTS

A. Data and local tracking

We use the released per-vehicle, no-fusion detections from DMSTrack [9] on V2V4Real [8]. Nine validation sequences provide the development subset; the main evaluation contains 25 complete train sequences. The two subsets contain 1,993 and 5,601 paired frames. All 34 sequences informed method development, and the released detector was trained on the train split; the evaluation therefore measures fusion performance on this development corpus.

We track planar car centers within 40 m of either sensor and the annotated ego-frame rectangle $[-70.4, 70.4] \times [-40, 40]$ m, excluding a 3 m disk around each vehicle. The same domain applies to truth and estimates. Each local sensor uses its own 40 m support, with transforms supplied by the dataset. Detections retain the released confidence cutoff of 0.20. The model uses current ego coordinates without ego-motion compensation.

All methods share the score-aware LBP-LMB local update. A constant-velocity model uses a 0.1 s interval, survival probability 0.99, acceleration intensity $25 \text{ m}^2/\text{s}^3$, position covariance $I_2 \text{ m}^2$, and in-domain $p_D = 0.9$. Poisson clutter has mean three per sensor scan and spatial density $3/(\pi 40^2)$. Each previous-scan detection creates an independent local birth with existence 0.01, zero velocity, and covariance $\text{diag}(16, 16, 225, 225)$. Tracks with $r \leq 0.001$ are pruned; retained mixtures are moment-projected to one Gaussian, and MAP cardinality determines set extraction. History qualification and absent-label censoring are shared across the geometric methods. A censor uses existence 0.001 when the represented mean lies in its sensor's support.

B. Detection-score likelihood

Per-sensor one-to-one assignment within 12 m supplies offline calibration labels. We fit $u(s) = \sigma(a \logit s + b)$, $a \geq 0$, by sequence-balanced binary log loss with regularization $0.001a^2$. Development sequences use leave-one-sequence-out fits. The main evaluation uses a fit on all nine development sequences, held fixed at $a = 2.8858$, $b = 2.1564$. The score enters each local association likelihood once through

$$L(s) = \frac{u(s)}{1 - u(s)} \frac{1 - \pi}{\pi}, \quad (21)$$

where π is the sequence-balanced fitting prevalence. This separable mark model is common to GCE and every main baseline. Online tracking and fusion use detections and communicated beliefs.

C. Communication and baselines

Reliable links deliver both directed packets each scan. Intermittent links combine independent directed loss of 0.1 with a bidirectional outage over the middle 20% of scans. Methods share the delivery trace and one packet opportunity per direction per scan. A receiver pools its own and a delivered neighbor's update with weights $[0.5, 0.5]$; without a neighbor packet it keeps its local update. We account for 16,384-byte fragments and a 256-byte control charge per

scan. Delivery is independent of packet length. GCE uses $\rho = 0.25$ and $\lambda = 5 \text{ s}$.

No-age KLA uses qualified weights α, b . *Recency* replaces b with q in existence, while *Conservative* uses β without a current correction. *Capped-score* and *Capped-cal.* bound recency existence by $\min(r_{\text{rec}}, \max(r_{\text{noage}}, U))$; U is the largest association-weighted raw or calibrated score among currently observed, upweighted sources with existence at least 0.5. *Scalar* uses GCE's base and admission gates to correct existence log-odds, retaining the original spatial pool and normalizer without a Gaussian ratio or curvature test.

MIL-AM adapts the public manuscript's augmented label assignment and common/exclusive subspace fusion [14]. Symmetric KL cost with unmatched cost 50 per side aligns Gaussian labels; the geometric methods use the same alignment. Common labels use arithmetic existence pooling and existence-weighted spatial mixtures; exclusive labels retain their source density. *TC-5/TC-10* use the author's kinematic matching and two-stage multi-neighbor functions [13], with five- or ten-scan histories, matching order one, and cutoff 12 m. They combine independently labeled local tracks and retain the architecture's separate local filters. *Local* uses the shared backend without communication.

D. Metrics and ablations

We report position OSPA with $p = 2, c = 12 \text{ m}$ [17] and the missed and false squared-cost components of GOSPA with $p = 2, c = 12 \text{ m}, \alpha = 2$ [18]. Common-target RMSE pools errors on robot-time-truth instances matched by both compared methods. Sequences receive equal weight; 95% percentile intervals use 10,000 paired resamples of complete sequences.

Three ablations remove the curvature guard, conservative history switch, or positive score constraint, one at a time, and run the complete recursive tracker. The score ablation retains the common local score model. *Scalar* provides a separate reference for existence-only correction. Analytical checks cover shared-prior fusion, conflicting observations, zero gates, single-source updates, and nonintegrable aggregates. Codec evaluation uses all 34 sequences under both link conditions and compares decoded values and tracking trajectories.

V. RESULTS

A. Lower set error across both link conditions

GCE achieves the lowest OSPA among all main comparison methods (Table I). It improves on No-age KLA by 6.7%/5.7% and on Recency by 5.9%/4.8% (reliable/intermittent). Figure 2 shows that the paired OSPA intervals lie below zero against every geometric reference, including both capped variants and *Scalar*. The improvement is thus consistent across several ways of weighting inherited posteriors.

The gain improves several aspects of tracking together. Relative to No-age KLA, missed-target cost falls by 16.9%/13.6% and false-target cost by 10.8%/6.3%. Common-target RMSE also drops from 0.964 to 0.907 m

TABLE I

MAIN COMPARISON ON 25 SEQUENCES WITH A SHARED LOCAL TRACKER. OSPA IS IN M; MISSED AND FALSE GOSPA COSTS ARE IN M^2 . LOWER IS BETTER; BEST VALUES ARE BOLD.

Method	Reliable links			Intermittent links		
	OSPA	Missed	False	OSPA	Missed	False
Local	5.914	190.030	45.556	5.914	190.030	45.556
MIL-AM	4.014	74.566	56.973	4.163	82.428	58.363
TC-5	4.253	68.620	75.289	4.693	101.053	64.733
TC-10	4.414	88.338	65.001	4.829	116.571	57.691
No-age KLA	3.702	85.564	40.715	3.988	92.977	44.776
Recency	3.672	78.330	43.986	3.952	87.113	47.603
Conservative	3.672	86.046	37.436	3.956	93.865	42.483
Capped-score	3.666	83.065	39.074	3.913	90.189	43.663
Capped-cal.	3.650	80.776	40.418	3.897	88.391	44.318
Scalar	3.526	78.625	34.536	3.832	86.342	40.523
GCE	3.456	71.095	36.303	3.761	80.344	41.971

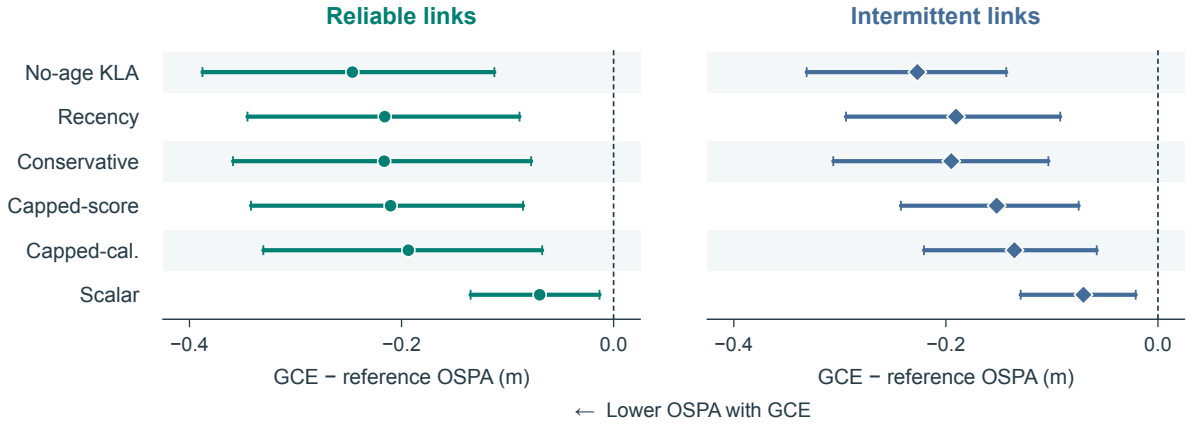


Fig. 2. GCE improves on each geometric reference under both link conditions. Points show equal-sequence mean OSPA differences; bars are paired 95% percentile intervals over the 25 sequences. All intervals lie to the left of zero, favoring GCE.

under reliable links and from 1.085 to 1.050 m under intermittent links. Recency exhibits the same direction of improvement across these metrics. GCE therefore improves target recovery and localization while suppressing false tracks. Its OSPA also remains below the arithmetic MIL-AM and both track-consensus adaptations.

B. Joint correction and component ablations

Table II separates the Scalar reference, three component removals, and the complete GCE rule. GCE improves on Scalar in both conditions, with both paired OSPA intervals below zero (Fig. 2). It reduces missed-target cost and common-target localization error, with a small increase in false-target cost. Applying the joint correction to the same saved fusion inputs also lowers OSPA relative to the scalar substitution. The immediate fusion benefit therefore accompanies the improvement accumulated through recursive tracking.

The curvature guard lowers mean OSPA in both conditions, with a paired interval below zero under intermittent links. It also suppresses the large position shifts produced by

unsuitable Gaussian ratios; removing it activates the aggregate fallback twice. The conservative history switch makes a smaller contribution. Removing the positive score constraint reduces missed-target cost but substantially increases false-target cost, showing its role in controlling positive evidence. For the history and score removals, paired total-OSPA intervals include zero. Together, the results support the complete joint correction over Scalar and show how the positive score constraint controls false tracks.

The nine-sequence development subset shows the same mean OSPA ordering of GCE, Scalar, and No-age KLA. Detection-score modeling also improves the local backend: unmarked No-age KLA is weaker than its score-aware counterpart. Because every main method shares the latter, the main comparison measures the additional improvement from fusion.

C. Communication-accuracy tradeoff

The zero-vector codec preserves the full Gaussian GCE trajectories while reducing raw payload by 20.3%/21.1%. Figure 3 shows the resulting shift toward lower communication at unchanged OSPA. With fragmentation and control

TABLE II
ABLATIONS OF THE RECURSIVE TRACKER. EACH W/O ROW REMOVES ONE GCE COMPONENT; SCALAR IS A SEPARATE EXISTENCE-ONLY REFERENCE. UNITS AND BOLDFACE FOLLOW TABLE I.

Method	Reliable links			Intermittent links		
	OSPA	Missed	False	OSPA	Missed	False
Scalar reference	3.526	78.625	34.536	3.832	86.342	40.523
w/o curvature guard	3.491	74.845	35.376	3.805	84.047	41.615
w/o history switch	3.467	70.282	38.148	3.776	80.266	43.105
w/o score constraint	3.713	55.993	59.014	3.922	69.042	57.784
GCE (complete)	3.456	71.095	36.303	3.761	80.344	41.971

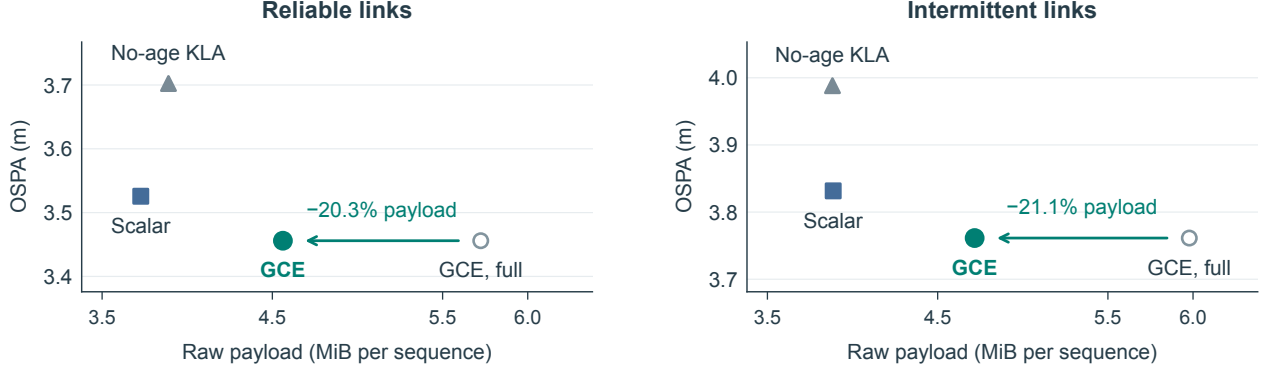


Fig. 3. Communication and tracking accuracy, averaged over 25 sequences. The arrows connect full and encoded GCE packets: exact zero omission lowers raw payload while retaining identical OSPA. No-age KLA and Scalar provide the reference tradeoffs. Lower and leftward are better.

TABLE III
MEAN COMMUNICATION PER SEQUENCE (MiB). FRAG. INCLUDES PACKET FRAGMENTATION AND CONTROL TRAFFIC.

Method	Reliable links		Intermittent links	
	Raw	Frag.	Raw	Frag.
No-age KLA	3.89	8.35	3.88	8.29
Scalar	3.73	8.19	3.89	8.32
GCE (full)	5.72	9.79	5.98	9.95
GCE (encoded)	4.56	8.69	4.72	8.88

traffic, the reduction is 11.3%/10.8% (Table III). Encoded GCE requires 4.1%/7.0% more fragmented bytes than No-age KLA, in exchange for its lower tracking error. The codec thus recovers much of the cost of joint Gaussian evidence without weakening the fusion rule.

VI. DISCUSSION AND CONCLUSION

GCE restores bounded current evidence on top of an inherited posterior pool and normalizes the resulting Bernoulli jointly. This coupling is central: a spatial update changes both target localization and the singleton mass that determines existence. The Gaussian form supplies closed-form fusion, while source admission controls approximate local increments. Across both communication conditions, the complete rule improves on the main fusion references and combines

lower missed and false costs with better localization relative to no-age and recency pooling. Exact zero omission makes the additional evidence cheaper to communicate at identical tracking output.

The present evaluation uses approximate associations, a fitted score model, planar two-vehicle tracking, and emulated links. A curvature test ensures admissible Gaussian correction but does not resolve unknown cross-source correlation. Evaluation on new routes and detector splits, larger robot teams, and measured radio links will test transfer beyond the current corpus. The main finding is that current evidence is most useful when its spatial and existence effects are retained together.

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